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Studierichting: Informatica
Oefeningenreeks 5A nr. 4.17

$$BI \int_0^1 \frac{dx}{5x^2+5} \Rightarrow OBI = \int \frac{1}{5x^2+5} dx$$
$$= \int \frac{1}{5\left(x^2+1\right)} dx = \frac{1}{5} \int \frac{1}{x^2+1} dx = \frac{1}{5} \operatorname{Bgtan}(x) + c$$
$$BI = \left[\frac{1}{5} \operatorname{Bgtan}(x) + c\right]_0^1 = \left(\frac{1}{5} \operatorname{Bgtan}(1) + c\right) - \left(\frac{1}{5} \operatorname{Bgtan}(x) + c\right) = \frac{1}{5} \cdot$$
$$\frac{\pi}{4} \cancel{+c} - \cancel{\frac{1}{5} \cdot 0} \cancel{-c} = \frac{\pi}{20}$$

References